SPAGHETTILAB | ONE BOARD, MANY POSSIBILITIES

Connect a circuit, choose how each port behaves, and explore what happens.

CONNECTIONS & POWER
OPTIONAL ESP32 MODULE
ESP32-S3 | Optional wireless module
Adds Wi-Fi, Bluetooth and ESP32 project support.
Adds Wi-Fi, Bluetooth and support for ESP32 projects.
U1
ESP32-S3-WROOM-1_1U_FlexyPin_Socket
DIRECT ACCESS
J3
M20-7810645R
Connects accessories directly to the optional ESP32.
MAIN POWER
U4
TPS562201DDCR
TPS562201 | Main power supply
Creates the power used by the board.
Turns the incoming power into the supply used by the board.
PORT LIGHTS
SK6812MINI-E
Shows the state of each port with coloured light.
MEASURE | INTERNAL CONVERTERS
Six port voltages -> AG32 ADC
0–3.3 V, common ground; calibrate before use.
OPTIONAL ACCESSORY LINKS
Provides the supply voltages used by supported accessories.
QWIIC ACCESSORIES
SM04B-SRSS-TB(LF)(SN)
Connects compatible sensors with a small cable.
LIVE VIEW | SIX PORT VOLTAGES
Native ADC inputs remain on PORT_A0–A5.
Current is measured separately by INA226.
No external ADC, DAC, reference or op-amps.
Connects displays, cameras and other supported add-ons.
BOARD-TO-BOARD LINK
FROM_PREVIOUS
TO NEXT
AO3401A
Q2
U8
G3VM-61VY3
G3VM | Network end switch
Enables the end-of-network connection when needed.
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
Passes power and communication to neighbouring boards.
NETWORK CONNECTION
U2
SN65HVD230DR
SN65HVD230 | Network interface
Connects this board to the shared CAN network.
Lets this board exchange messages with the other boards.
USB CONNECTION
U3
HC-TYPE-C-16P-01A
Connects the board to a computer.
OPTION | ESP32-ONLY CONTROLLER
Uses the ESP32 to configure the ports when AG32 is not fitted.
SIGNAL CONNECTIONS
U12
U13
SN74CBTLV3125PWR
U12 / U13 | Bidirectional signal switches
Connect or disconnect the ESP32 from the experiment ports.
Connects or separates the ESP32 signals from the experiment.
Existing circuit preserved. Tester features require firmware and prototype validation; no new tester components were added.
SIX EXPERIMENT PORTS
AO6604
J1 M20-7810645R
J140 M20-7810645R
Gives each port a signal, power or ground connection.
Q3–Q8 | Electronic power switches
Give a port power or ground when selected.
CONTROL & LAB TOOLS
U14A–U14G are seven sections of ONE physical AG32 chip.
PORT SUPERVISOR
U14
AG32VF407VGT6
AG32 section A | Experiment signals
Reads the ports and handles digital signals.
Observes the ports and manages digital signals for the experiment.
PORT MODE CONTROL
AG32 section B | Port settings
Chooses signal, power or ground for each port.
Chooses how each experiment port is connected.
SUPERVISOR START-UP
X1
20MHz
20 MHz | Controller clock
Keeps the controller running at a steady pace.
AG32 section F | Power and start-up
Starts the controller and provides programming access.
Powers, starts and provides access to the board controller.
COMMUNICATION HUB
AG32 section C | Board communication
Exchanges data with the computer and other boards.
Links the controller to USB, the network and the optional ESP32.
CURRENT MEASUREMENT
AG32 section D | Current scan control
Chooses which port is measured.
74HC4051BQ,115
4051 | First measurement selector
Connects one side of the selected current sensor path.
4051 | Second measurement selector
Connects the other side of the same sensor path.
U16
INA226AIDGSR
INA226 | Current sensor
Measures the current of the selected port.
Selects one port at a time and measures the current flowing through it.
LOCAL POWER SUPPORT
Keeps the controller supply steady while it works.
SCAN | SMALL TEST SIGNALS
U19
74HC4051BQ,115
R77
4.7k 0.1% 25ppm
Internal DAC -> R77 -> selectable port. Read both sides of R77 with internal ADC.
Scan: MOSFETs OFF, digital drivers Hi-Z, ESP switches OPEN, R77 bypass OFF.
A return path is required. Occupancy detection is assisted, not guaranteed.
Applies a gentle test signal and watches how each connection responds.
ESP32 PROGRAMS THE AG32
Loads the board controller without an external programmer.
GPIO47 TX / GPIO48 RX | GPIO15 BOOT0 | GPIO16 RESET (open-drain)
AG UART0: TX pin68 / RX pin69. BOOT1 stays LOW via R84.
Recovery: keep ESP32 pins Hi-Z; use UART pads or the existing JTAG pads.
AG Base: U14 fitted, Lab components omitted, R56–R61 = 0R.
ESP Base: U25 + R98–R102 fitted; U14, Lab and AG programming parts omitted.
Lab: U14 + measurement components fitted. USB remains direct to ESP32.

Single-page schematic. Functional descriptions are not performance specifications.

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